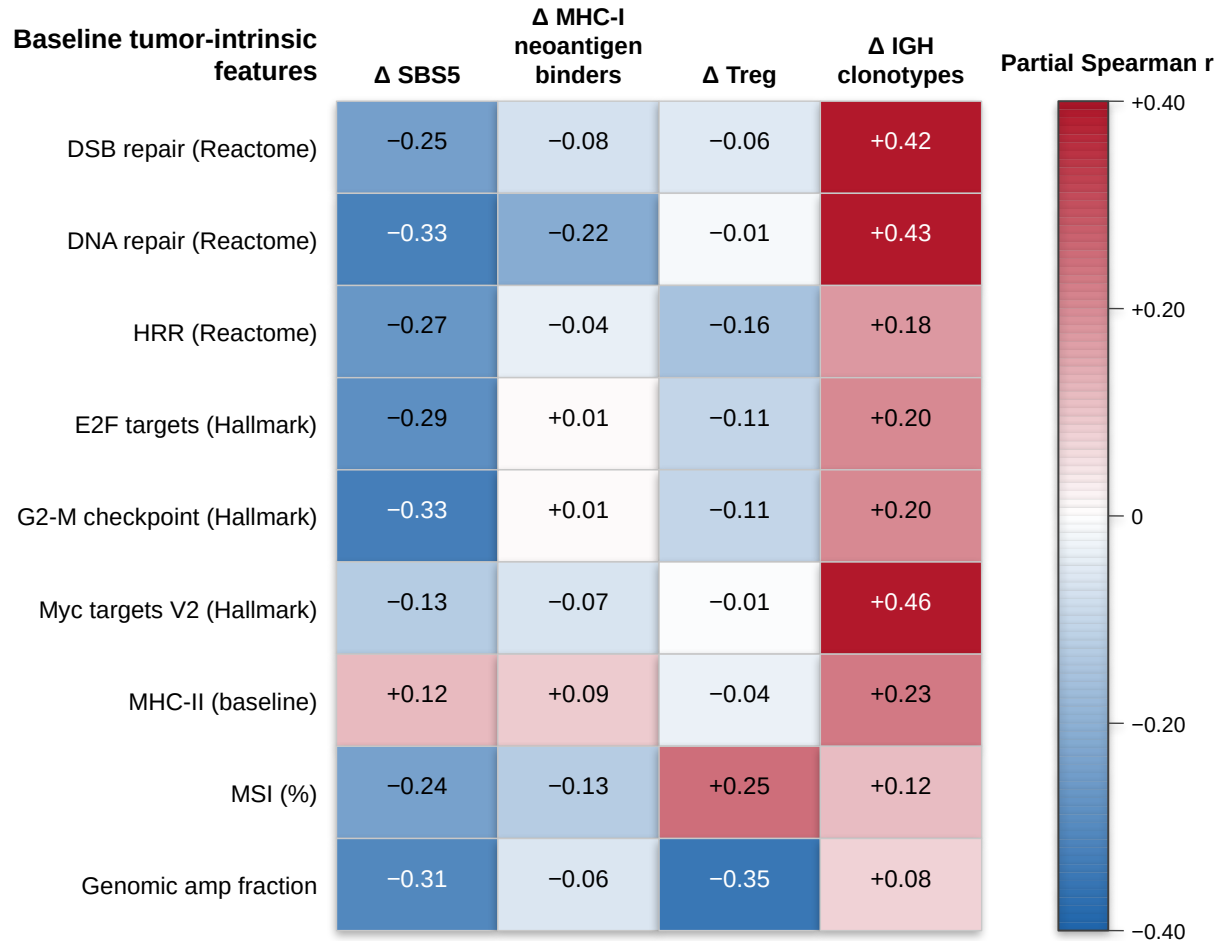


### Cascade $\Delta$ features (radiation-phase dynamics)



#### Sensitivity analysis — partial Spearman with response-group adjustment.

Partial Spearman  $\rho$  · BH-corrected across 36 pairs. P value shown for cells with  $P < 0.10$ .

Partial  $r$  range [-0.48, +0.46] (plain Spearman range [-0.54, +0.56] compressed by group adjustment).

0/36 partial  $P < 0.05$  · 0/36 BH  $q < 0.05$  (robust to group adjustment).

Caveat: Conditioning on response group (an outcome variable in our analytical design) may introduce collider bias under some causal structures. This sensitivity